An Elementary Uniform Spectral Gap for the Syracuse Transfer Operator

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https://github.com/m4cd4r4/collatz-conjecture-spectral-gap

July 2026

Abstract

Let T_k be the transfer operator of the Syracuse map $\mathrm{Syr}(n) = (3n+1)/2^{v_2(3n+1)}$ on the $N = 2^{k-1}$ odd residues modulo 2^k, averaging over the 2^k lifts of a residue. We prove a uniform spectral gap: for every $k \geq 3$,

$$|\lambda_2(T_k)| \;\leq\; \mathrm{cert}(k) \;\leq\; 2^{-3/2} + 2^{-1} \;=\; 0.8535\ldots \;<\; 1,$$

where $\mathrm{cert}(k)$ is an explicit weighted row-sum certificate over a 2-adic frequency grading. The proof is entirely elementary: finite cyclic 2-group arithmetic (three uniform-fiber lemmas sharing one engine, "3 is a unit modulo 2^k"), a finite geometric series, an integer collision count, and a single application of the Cauchy–Schwarz inequality. A sharper per-level bound on the defect covector improves the constant to 0.6827; a further sharpening to 0.6553 rests on a constant verified only for $k \leq 26$ and is labelled as such throughout. The measured values are $\mathrm{cert}(k) \approx 0.6345$ and $|\lambda_2| \approx 0.27$. The elementary core of the proof is machine-checked in Lean 4 (Mathlib, sorry-free).

What it measures. The honest 2-adic kernel, conditioned on the full fibre rather than on a finite lift window, is nilpotent off its Perron eigenvalue, so the mixing of the averaged mod-2^k dynamics is trivial. T_k differs from it by a rank-one but *not small* defect ($\|T_k - K_k\|_1 \approx 0.4$), and the theorem is the statement that this large rank-one truncation defect cannot drive the spectrum to the unit circle at any scale — a uniform stability statement for the finite-window approximation.

Scope. This is a theorem about the Markov model, not about Collatz orbits. In particular it does *not* rule out Collatz cycles: the analogous $3x-1$ operator passes the identical certificate even more strongly, yet $3x-1$ has non-trivial cycles. An earlier version of this project claimed a cycle-elimination corollary; that claim was withdrawn (Section 9).

Keywords: Syracuse map, transfer operator, spectral gap, 2-adic analysis.

*With Claude (Anthropic): Opus 4.8 research sessions and a Fable 5 adversarial review, refereeing, and consolidation pass. All proofs are elementary and every numerical claim is reproducible from the repository.

1 Introduction

For odd n, the Syracuse map is $\mathrm{Syr}(n) = (3n+1)/2^{v_2(3n+1)}$, where v_2 is the 2-adic valuation. Reducing modulo 2^k gives a natural Markov chain on the $N = 2^{k-1}$ odd residues: from a residue r, move to $\mathrm{Syr}(r + m2^k) \bmod 2^k$ with m uniform on $[0, 2^k)$. Its transition matrix T_k (column-stochastic) is the finite-scale transfer operator of the map; the mixing of this family is the standard probabilistic model for the equidistribution of Syracuse trajectories modulo powers of two, going back to the heuristics surveyed in [1] and quantified in Tao's almost-all result [2].

Our main result is a uniform, explicit, and elementarily proved spectral gap for this family.

Theorem 1.1 (Uniform spectral gap)**.** *For every $k \geq 3$,*

$$|\lambda_2(T_k)| \;\leq\; \mathrm{cert}(k) \;:=\; \max_{0\leq a\leq k-2} \sum_{b=0}^{k-2} Q_k[a,b]\, 2^{a-b} \;\leq\; 2^{-3/2} + 2^{-1} \;<\; 1,$$

where λ_2 is the second-largest eigenvalue in modulus and $Q_k[a,b] = \|P_a U_k P_b\|_2$ is the matrix of block operator norms of the Koopman adjoint $U_k = T_k^\top$ over the 2-adic frequency grading of Section 2.

What the theorem says, informally

A reader who wants the shape of the result before the notation may take the following as the whole story.

The Collatz map is deterministic, and deterministic questions about it are hard. A standard softening is to stop following one number and follow instead *where numbers land on average*, recording only the remainder modulo a power of two. That turns the map into a random walk on a finite set of states — the 2^{k-1} odd residues mod 2^k — and T_k is the matrix of that walk. The averaging is over the 2^k integers sharing a given residue.

A random walk "mixes" when the memory of where it started decays. The rate is governed by the second-largest eigenvalue λ_2: mixing happens exactly when $|\lambda_2| < 1$, and faster the smaller it is. For a *single* k this is a finite computation and carries no information. The content of Theorem 1.1 is that one constant works for *every* k at once, even though the matrix grows exponentially.

The proof has one idea, applied three times. Multiplication by 3 is invertible modulo any power of two — because 3 is odd — so it permutes residues and spreads any set of them perfectly evenly across the classes it can reach. Everything else is bookkeeping: sorting residues by how divisible they are by two, summing a geometric series over those layers, counting collisions in one exceptional row, and applying Cauchy–Schwarz once. There is no analysis and no computer search inside the proof.

The one place the argument could have failed, and does not, is that top layer. The cascade of blocks *truncates* there rather than continuing, and it is exactly that truncation which makes an L^2 estimate strong enough. Without it the bound would exceed 1 and say nothing.

Finally, and this is not a caveat but the main thing to carry away: none of this bears on the Collatz conjecture, and Section 9 shows by an explicit control that it cannot. Averaging destroys the very structure a cycle lives in.

Why the theorem is of interest

The interest of the theorem is threefold. First, uniformity: the bound is independent of k, while the state space grows as 2^{k-1}. Second, the constant is explicit and the true values sit far below it ($|\lambda_2| \approx 0.27$ for all tested k), so the certificate has a large margin. Third, and mainly, the proof is elementary from end to end. What looks like it should require Gauss-sum estimates reduces to three instances of one fact (multiplication by a unit has uniform fibers on a cyclic 2-group), one finite geometric series, one integer collision count with a two-line injectivity argument, and one Cauchy–Schwarz application. No analytic number theory, no perturbation theory, and no computer-assisted spectral certification enters the proof; numerics appear only as consistency checks.

We are explicit about history because this project previously overclaimed twice, and both corrections shaped the final form. (i) A March 2026 draft asserted the gap via an eigenvalue perturbation telescope with a "bounded condition number"; an adversarial audit (Session 39) refuted this: the eigenvector condition number grows like 10^k, and the published inductive inequality fails numerically at every odd $k \geq 7$. The present proof replaces eigenvalue perturbation with norm inequalities that are condition-number-free. (ii) The certificate was then presented as eliminating non-trivial Collatz cycles. That inference is false and was withdrawn; Section 9 records the decisive control experiment. What survives is the operator theorem itself.

Proof architecture

1. **Spectral reduction** (Section 3): $|\lambda_2(T_k)| \leq \rho(Q_k) \leq \mathrm{cert}(k)$, by a one-sided Perron split, a level-energy majorisation applied to a compression, and Gelfand's formula. A subtlety repaired late in the project: T_k is *not* doubly stochastic and its stationary distribution is *not* exactly uniform, so the argument must (and does) avoid both.

2. **Block structure** (Section 4): $U_k = U_{\mathrm{clean}} + D$ where U_{clean} is strictly upper-triangular in the level grading with exactly computable block norms $\|P_a U_{\mathrm{clean}} P_b\| = 2^{-(b-a)/2}$ (Lemma A), and D is a rank-1 defect supported at the single residue $r^* = -3^{-1} \bmod 2^k$.

3. **Defect mass** (Section 5): the defect covector c satisfies $\|c\|_2 \leq \sqrt{3}\, 2^{-k/2}$, by an integer collision count (Lemma B).

4. **Assembly** (Section 6): each weighted row sum is bounded by the envelope $f(e) = \sum_{d=1}^{e-2} 2^{-3d/2} + 2^{-(e-1)/2}$ with $e = k - a$, whose maximum is $f(3) = 2^{-3/2} + 2^{-1}$. The only analytic input is Cauchy–Schwarz; crucially, the upper cascade *truncates* at the top rows, which is what makes the L^2 bound sufficient.

Section 7 records the sharper per-level defect bound (Lemma C, via an elementary frequency-doubling identity), which improves the constant to 0.6827 but is not needed for Theorem 1.1. Its sharp form gives 0.6553 but consumes a constant verified only to $k \leq 26$; we keep the two apart and never quote the sharper number as proved. Section 8 describes the Lean 4 formalisation of the elementary core. Section 9 delimits scope.

2 The operator and the frequency grading

Definition 2.1. *Fix $k \geq 3$ and let $N = 2^{k-1}$. The state space is the set of odd residues modulo 2^k. The chain moves from r to $\mathrm{Syr}(r + m2^k) \bmod 2^k$ with m uniform on $\{0, \dots, 2^k - 1\}$; $T_k[s', s]$ is this transition probability, and $U = U_k := T_k^\top$ is the Koopman (expectation) operator, $(Ug)(r) = \mathbb{E}_m\, g\big(\mathrm{Syr}(r + m2^k)\big)$.*

T_k is column-stochastic by construction, so $\mathbf{1}^\top T_k = \mathbf{1}^\top$ and $\rho(T_k) = 1$.

Remark 2.2 (T_k is not doubly stochastic)**.** *The row sums of T_k are* not *identically 1: numerically $\|T_k\mathbf{1} - \mathbf{1}\|_2 \approx 0.764 \cdot 2^{-k/2}$, another face of the r^* defect of Section 4. Consequently the stationary distribution is only approximately uniform, U does not preserve the mean-zero subspace, and identities such as $T_r^p = T_k^p - N^{-1}J$ (used in an earlier write-up) are false. Nothing below uses uniformity of the stationary distribution; the reduction in Section 3 is deliberately one-sided.*

Definition 2.3 (Characters and levels)**.** *Let $\omega = e^{2\pi i/2^k}$ and, for $\xi \in [0, 2^{k-1})$, let $\chi_\xi(r) = \omega^{\xi r}/\sqrt{N}$ on odd r. On odd arguments $\chi_{\xi+2^{k-1}} = -\chi_\xi$, so this range selects one representative per proportionality class, and the χ_ξ form an orthonormal basis of $\mathbb{C}^N$ (orthogonality is the $m = k$ case of Lemma 4.3 below). χ_0 is the constant (Perron) vector. The* level *of $\xi \neq 0$ is $v_2(\xi) \in \{0, \dots, k-2\}$; P_a denotes the orthogonal projection onto $W_a = \mathrm{span}\{\chi_\xi : v_2(\xi) = a\}$, of dimension $d_a = 2^{k-2-a}$. The mean-zero space is $V = W_0 \oplus \cdots \oplus W_{k-2}$, an orthogonal decomposition.*

3 The spectral reduction

Proposition 3.1. *Let $Q = Q_k$ with $Q[a,b] = \|P_a U P_b\|_2$. Then $|\lambda_2(T_k)| \leq \rho(Q) \leq \mathrm{cert}(k)$.*

Proof. (i) Perron split. Since $\mathbf{1}^\top T = \mathbf{1}^\top$, the mean-zero space $V = \ker \mathbf{1}^\top = \mathbf{1}^\perp$ is T-invariant. In a basis adapted to $\mathbb{C}^N = \mathrm{span}(\mathbf{1}) \oplus V$, T is block-triangular $\left(\begin{smallmatrix} 1 & 0 \\ d & A \end{smallmatrix}\right)$ with $A = T|_V$ (the vector d, the V-component of $T\mathbf{1}$, is nonzero by Remark 2.2 but harmless). The characteristic polynomial factors, so $\mathrm{spec}(T) = \{1\} \uplus \mathrm{spec}(A)$ and $|\lambda_2(T)| \leq \rho(A)$.

(ii) Adjoint and Gelfand. $A^* = (P_V T P_V)^*|_V = P_V U P_V =: U_V$, the compression of U to V (V need not be U-invariant; only the compression is used), and $\rho(A) = \rho(U_V)$. Gelfand's formula $\rho(U_V) \leq \|U_V^p\|_2^{1/p}$ holds for every matrix and every p; no normality or diagonalisability enters. This is the step that makes the argument immune to the 10^k eigenvector condition numbers that invalidated the earlier perturbative route.

(iii) Level majorisation. For $g \in V$ let $\alpha(g)_a := \|P_a g\|_2$. Since $P_a P_V = P_a$, the triangle inequality gives, for each level a',

$$\|P_{a'} U_V g\| = \Big\| \sum_b P_{a'} U P_b g \Big\| \leq \sum_b Q[a', b]\, \alpha(g)_b = (Q\, \alpha(g))_{a'},$$

i.e. $\alpha(U_V g) \leq Q\, \alpha(g)$ entrywise. As $Q \geq 0$ this iterates: $\alpha(U_V^p g) \leq Q^p \alpha(g)$. The levels are orthogonal and exhaust V, so $\|U_V^p g\|_2 = \|\alpha(U_V^p g)\|_2 \leq \|Q^p\|_2 \|g\|_2$, whence $\rho(U_V) \leq \lim_p \|Q^p\|^{1/p} = \rho(Q)$.

(iv) Weighted row sum. For the diagonal similarity $S = \mathrm{diag}(2^0, \dots, 2^{k-2})$ and the nonnegative matrix Q, $\rho(Q) = \rho(SQS^{-1}) \leq \|SQS^{-1}\|_\infty = \max_a \sum_b Q[a,b]\, 2^{a-b} = \mathrm{cert}(k)$. ◻

4 Block structure: three instances of one engine

Throughout, x is odd, $v(r) := v_2(3r+1) \geq 1$, and 3^{-1} denotes the inverse of 3 in the relevant $\mathbb{Z}/2^m$.

4.1 Coset uniformity and the splitting $U = U_{\text{clean}} + D$

Lemma 4.1 (CU)**.** *Let x be odd with $v := v_2(3x+1) < K$, and $q_0 := (3x+1)/2^v$ (odd). Then for every m,*

$$v_2\Big(3(x+m2^K)+1\Big) = v, \qquad \mathrm{Syr}(x+m2^K) = q_0 + 3m\,2^{K-v},$$

and as m ranges over $\mathbb{Z}/2^K$ the residue $\mathrm{Syr}(x+m2^K) \bmod 2^K$ is uniform on the coset $q_0 + 2^{K-v}(\mathbb{Z}/2^K)$ of size 2^v, each value attained exactly 2^{K-v} times.

Proof. $3(x+m2^K)+1 = 2^v q_0 + 3m2^K = 2^v(q_0+3m\,2^{K-v})$, and the bracket is odd because q_0 is odd and $K-v \geq 1$; this freezes the valuation and gives the affine formula. For the count: $m \mapsto 3 \cdot 2^{K-v} m$ is an endomorphism of $\mathbb{Z}/2^K$ whose image is the subgroup $2^{K-v}(\mathbb{Z}/2^K)$ of order 2^v (3 is a unit) and whose kernel has order 2^{K-v}; every image point has exactly 2^{K-v} preimages, and adding q_0 translates the image to the coset. □

Apply Lemma 4.1 with $K = k$. For every row r with $v(r) < k$, averaging χ_η over the uniform coset collapses to a geometric sum which is 2^v or 0 according to whether $2^v \mid \eta$, yielding the masked-phase formula

$$(U\chi_\eta)(r) = \mathbf{1}[\,v(r) \leq v_2(\eta)\,]\;\omega^{\eta\, q(r)}, \qquad q(r) := (3r+1)/2^{v(r)}, \qquad (v(r) < k). \tag{1}$$

Exactly one odd residue fails the hypothesis: $2^k \mid 3r+1$ has the unique odd solution $r^* = -3^{-1} \bmod 2^k = (4^{\lceil k/2\rceil}-1)/3$, where $3r^*+1 = 2^{2\lceil k/2\rceil}$ exactly. Define U_{clean} by (1) with the r^*-row set to zero, and $D := U - U_{\text{clean}}$. Then:

- **(R1)** $P_a U_{\text{clean}} P_b = 0$ for $a \geq b$ (strict upper-triangularity; a by-product of the survivor analysis below);
- **(R2)** $D = e_{r^*} c^*$ is rank 1, where c is the probability vector $c[t] = 2^{-k}\#\{m : \mathrm{Syr}(r^* + m2^k) \equiv t \bmod 2^k\}$; hence $\|P_a D P_b\| = u_a v_b$ with $u_a := \|P_a e_{r^*}\|$, $v_b := \|P_b c\|$;
- **(R3)** $u_a = 2^{-(a+1)/2}$ exactly: any state unit vector has flat character energy $1/N$ per frequency, so $u_a^2 = d_a/N = 2^{-1-a}$.

By the triangle inequality, for all a, b:

$$Q[a,b] \;\leq\; \|P_a U_{\text{clean}} P_b\| + u_a v_b. \tag{2}$$

4.2 The two remaining elementary inputs for Lemma A

Lemma 4.2 (SB: shell bijection)**.** *Fix $1 \leq j \leq k-2$. The map $\psi(r) := (3r+1)/2^j \bmod 2^{k-j}$ is a bijection from the shell $S_j := \{r \text{ odd}, 0 < r < 2^k,\ v(r) = j\}$ onto the odd residues modulo 2^{k-j}.*

Proof. Injectivity: $\psi(r) = \psi(r')$ gives $3r+1 \equiv 3r'+1 \pmod{2^k}$, so $r = r'$ (3 a unit, both in $[0, 2^k)$). Cardinality: $v(r) = j$ pins r to a single (odd) residue class modulo 2^{j+1}, so $|S_j| = 2^{k-j-1}$, which equals the number of odd residues modulo 2^{k-j}; injective plus equal finite cardinality gives bijective, with multiplicity exactly one. $\square$

Lemma 4.3 (S4: odd geometric sum). *For $m \geq 1$ and $S^{\text{odd}}(\alpha, m) := \sum_{q \text{ odd}, 0 \leq q < 2^m} \omega^{\alpha q}$: if $2^{k-1} \mid \alpha$ then $|S^{\text{odd}}| = 2^{m-1}$ (full); otherwise $S^{\text{odd}} = 0$ if and only if $k - m \leq v_2(\alpha) \leq k - 2$ (the dead band); below the band it is a nonzero partial sum.*

Proof. Substitute $q = 2\ell + 1$ and sum the geometric series with ratio $\omega^{2\alpha}$. $\square$

4.3 Lemma A: the clean upper cascade is a scaled isometry

Theorem 4.4 (Lemma A). *For all k and all $0 \leq a < b \leq k - 2$, with $d := b - a$, the block $B = P_a U_{\text{clean}} P_b$ satisfies $B^* B = 2^{-d} I$; in particular $\|P_a U_{\text{clean}} P_b\|_2 = 2^{-d/2}$ exactly. Moreover $P_a U_{\text{clean}} P_b = 0$ for $a \geq b$ (R1).*

Proof (survivor analysis). The matrix entry of U_{clean} between source χ_η ($v_2(\eta) = b$) and target χ_ξ ($v_2(\xi) = a$) is, up to the $1/N$ normalisation, $\hat{U}(\eta, \xi) = \sum_{r, 1 \leq v(r) \leq b} \omega^{\eta q(r) - \xi r}$. Group by the shell $j = v(r)$. On shell j, $r = 3^{-1}(2^j q - 1)$ with $q = q(r)$, so the phase is $q\,\alpha_j + \xi 3^{-1}$ with $\alpha_j := \eta - \xi\, 2^j 3^{-1}$. The valuations are

$$v_2(\alpha_j) = a + j \ (j < d), \qquad v_2(\alpha_j) = b \ (j > d), \qquad v_2(\alpha_j) \geq b \ (j = d),$$

and in every case $v_2(\alpha_j) \geq j$, so $\omega^{q\alpha_j}$ depends only on $q \bmod 2^{k-j}$. By Lemma 4.2 the shell sum is exactly $\omega^{\xi 3^{-1}} S^{\text{odd}}(\alpha_j, k - j)$, and by Lemma 4.3 every shell lands in the dead band $[j, k-2]$ *except* possibly $j = d$, which survives if and only if $\alpha_d \in \{0, 2^{k-1}\}$, with full modulus 2^{k-d-1}. Counting: each η has exactly 2^d owners ξ (two residue classes modulo 2^{k-d}, each contributing 2^{d-1} values of valuation a in $[0, 2^{k-1})$), giving the diagonal $(B^*B)[\eta, \eta] = N^{-2} \cdot 2^d \cdot (2^{k-d-1})^2 = 2^{-d}$; and for fixed ξ exactly one of the two candidate sources lies in the index range $[0, 2^{k-1})$, so distinct columns have disjoint row support and the off-diagonal entries vanish. For $a \geq b$ the same computation gives $v_2(\alpha_j) = b \in [j, k-2]$ for every shell $j \leq b$, so every shell is annihilated: this is (R1). $\square$

5 Lemma B: the defect mass, by an integer collision count

Theorem 5.1 (Lemma B). $\displaystyle\sum_{b=0}^{k-2} v_b^2 \;\leq\; \|c\|_2^2 \;=\; \frac{\text{coll}(k)}{4^k} \;\leq\; \frac{3 \cdot 2^k}{4^k},$ *i.e.* $\|v\|_2 \leq \sqrt{3}\, 2^{-k/2}$, *where* $\text{coll}(k) = \sum_t \text{cf}[t]^2$ *and* $\text{cf}[t] = \#\{m \in [0, 2^k) : \text{Syr}(r^* + m2^k) \equiv t \bmod 2^k\}$.

Proof sketch (full details in the repository). Since $3r^* + 1 = 2^{2\lceil k/2 \rceil}$, the fiber is $\text{Syr}(r^* + m2^k) = \text{oddpart}(a + 3m) \bmod 2^k$ with $a = 2^{2\lceil k/2 \rceil - k} \in \{1, 2\}$, an arithmetic progression of length 2^k in the integers. Decompose it by 2-adic shells $A_j = \{x = a + 3m : v_2(x) = j\}$, of sizes $|A_j| = 2^{k-1-j}$ ($j \leq k - 1$) plus one top atom.

FACT 1 (per-shell injectivity). On each shell, $x \mapsto (x/2^j) \bmod 2^k$ is injective: if $x = 2^j u$, $x' = 2^j u'$ with $u \equiv u' \pmod{2^k}$, then $3(m - m') = 2^j(u - u') \equiv 0 \pmod{2^{j+k}}$; as $\gcd(2, 3) = 1$ this forces $2^{j+k} \mid m - m'$, and $|m - m'| < 2^k$ gives $m = m'$. Hence the fiber count is 0/1 per shell, and $\text{coll} = \sum_{j,j'} |R_j \cap R_{j'}|$ over the shell images R_j.

The diagonal contributes exactly 2^k. The cross terms obey $\sum_{j<j'} |R_j \cap R_{j'}| \le \sum_{j' \ge 2}(j'-1)2^{k-1-j'} + (k-1) = 2^{k-1} - 1$ (the closed form includes the top atom's at most $k-1$ partners, covered by the truncation slack of the geometric series). Hence $\mathrm{coll} \le 2^k + 2(2^{k-1}-1) < 3 \cdot 2^k$. The sharp constant is $\mathrm{coll}/2^k \to 31/12$, comfortably inside 3. Finally $\sum_b v_b^2 = \|c\|^2 - 1/N \le \|c\|^2$, the dropped term being the Perron coefficient of the probability vector c. $\square$

6 The assembly

Proof of Theorem 1.1. Fix a row $a \le k-2$ and set $e := k - a \ge 2$. Write the weight as $2^{a-b} = 2^a(1/2)^b$. By (2), Theorem 4.4 and (R1),

$$\sum_b Q[a,b]\, 2^{a-b} \;\le\; \underbrace{\sum_{d=1}^{e-2} 2^{-d/2}\, 2^{-d}}_{\text{clean cascade, truncated}} \;+\; \underbrace{u_a 2^a \sum_{b=0}^{k-2} v_b\, 2^{-b}}_{\text{defect, all } b}.$$

Two remarks. The clean sum *truncates* at $d \le e-2$ because $b \le k-2$; at the top row $a = k-2$ it is empty. The defect is summed over *all* b, which absorbs its contribution to the upper blocks with no separate estimate.

For the defect factor, Cauchy–Schwarz against the geometric weight and Theorem 5.1 give, with no information about the profile of (v_b),

$$\sum_{b=0}^{k-2} v_b\, 2^{-b} \;\le\; \|v\|_2 \Big(\sum_{b \ge 0} 4^{-b}\Big)^{1/2} \;\le\; \sqrt{3}\, 2^{-k/2} \cdot \tfrac{2}{\sqrt{3}} \;=\; 2^{1-k/2},$$

and with $u_a 2^a = 2^{(a-1)/2}$ (R3) the defect term is at most $2^{(a-1)/2}\, 2^{1-k/2} = 2^{-(e-1)/2}$. Hence

$$\sum_b Q[a,b]\, 2^{a-b} \;\le\; f(e) \;:=\; \sum_{d=1}^{e-2} 2^{-3d/2} + 2^{-(e-1)/2}.$$

The difference $f(e+1) - f(e) = 2^{-(e-1)/2}\big(2^{-(e-1)} - (1 - 2^{-1/2})\big)$ is positive only at $e = 2$, so f increases from $f(2) = 2^{-1/2}$ to its maximum $f(3) = 2^{-3/2} + 2^{-1} = 0.8535\ldots$ and decreases thereafter toward $G_{\mathrm{up}} = (2^{3/2} - 1)^{-1} = 0.5469\ldots$. Therefore $\mathrm{cert}(k) \le f(3) < 1$ for every $k \ge 3$, and Proposition 3.1 completes the proof. $\square$

Remark 6.1 (Why truncation matters)**.** *Without the truncation one would bound the row sum by $G_{\mathrm{up}} + 2^{-1/2} = 1.25 > 1$ and wrongly conclude that the L^2 defect bound is insufficient (an earlier version of the assembly did exactly this and introduced the per-level Lemma C as a necessity). Row-wise, the two terms of f trade off: rows with a long cascade have a small defect weight and vice versa, with the worst case at $e = 3$.*

Remark 6.2 (Sharpness and measured values)**.** *The envelope is asymptotically sharp at the bottom row ($e = k$), where both the true row sum and $f(k)$ converge to G_{up}; the row that sets the constant, $e = 3$, has a wide margin (true value ≈ 0.63 against $f(3) = 0.854$). Ground-truth values (dense operator, both parities; reproducible with `fable_assembly_check.py`):*

k	*4*	*6*	*8*	*10*	*12*	*13*
$\mathrm{cert}(k)$	*0.6117*	*0.6338*	*0.6347*	*0.6345*	*0.6345*	*0.6344*
$\rho(Q_k)$	*0.5020*	*0.5535*	*0.5661*	*0.5673*	*0.5656*	*0.5644*
$\|c\|_2\, 2^{k/2}$	*1.6202*	*1.6105*	*1.6081*	*1.6075*	*1.6073*	*1.6072*

All row bounds $R_a \le f(k-a)$ hold with margin at every tested k, and the rank-1 factorisation $Q[a,b] = u_a v_b$ on $a \ge b$ is machine-exact.

7 The per-level sharpening (Lemma C)

Theorem 1.1 needs only the L^2 mass of the defect. The per-level structure is nevertheless elementary and sharp, and improves the constant.

Theorem 7.1 (Lemma C). *$v_b \le \frac{3}{4}\, 2^{-b}\, 2^{-k/2}$ for all k and $0 \le b \le k-2$.*

The proof (in the repository) rests on a frequency-doubling identity for the fiber exponential sum $S_k(\xi) = \sum_t \mathrm{cf}[t]\, e^{-2\pi i \xi t/2^k}$:

Lemma 7.2 (H: homometry). *For $k \ge 2$ and $2\xi \notin \{0, 2^{k-1}\} \bmod 2^k$: $S_k(2\xi) = S_{k-1}(\xi)$ exactly.*

Proof idea. Split the fiber sum by the parity of m. One parity makes $3m + a$ odd and contributes a geometric series that vanishes away from the two excluded frequencies; the other parity halves the scale and reproduces the $(k-1)$-fiber exactly (the parity flip swaps the offsets $a = 1 \leftrightarrow 2$, matching the parity of $k-1$). □

Iterating Lemma 7.2 reduces the level-b energy to a single scale $p = k - b$, where a Parseval identity converts it into the half-shift autocorrelation $\mathrm{coll}(p) - A_p$, which the shell decomposition of Theorem 5.1 bounds by $3 \cdot 2^{p-1} - 2$ (the $j = 0$ shell cancels identically from the autocorrelation). Feeding Theorem 7.1 through the truncated envelope of Section 6 replaces $2^{-(e-1)/2}$ by $2^{-(e+1)/2}$ and yields

$$\mathrm{cert}(k) \;\le\; \max_{e \ge 2} \left[\sum_{d=1}^{e-2} 2^{-3d/2} + 2^{-(e+1)/2} \right] \;=\; 0.6559\ldots \qquad (\text{maximum at } e = 4),$$

remarkably close to the measured 0.6345.

8 Lean formalisation

Theorem 1.1 is machine-checked in Lean 4 against Mathlib (v4.27.0), end to end and with no undischarged hypothesis. The endpoint is `gap_certificate_unconditional`: for every $k \ge 3$, every eigenvalue $\mu \ne 1$ of the operator T_k — *defined inside the formalisation* from the lift window, not axiomatised — satisfies $\|\mu\| < 0.853554$.

The project is 18 files and 449 theorem/lemma declarations, with no `sorry`, no `native_decide`, and every `#print axioms` set a subset of {`propext`, `Classical.choice`, `Quot.sound`}. (That check is a subset test, not a string match: a number of declarations legitimately use fewer axioms.) The correspondence with this paper is:

- T_k **itself** — `TransferOperator.lean`. `Tcount k u r` counts the lifts $m < 2^k$ of the odd residue r with $\mathrm{Syr}(r+m2^k) \equiv u$, which is Section 2's definition transcribed. Nothing downstream is about an abstract operator satisfying axioms.

- **Proposition 3.1** (the spectral reduction) — `OperatorChain.lean`: the Perron split on $\ker \mathbf{1}^\top$, the compression, and the Gelfand step. The one-sided form is load-bearing and is the form proved: T_k is *not* doubly stochastic and its stationary vector is *not* uniform.

- **Lemma 4.1** (coset uniformity), including the fiber count, and the counting halves of Lemma 4.2 — `CountingLemmas.lean`.

- **Theorem 5.1** (the collision count) — `CollisionBound.lean`, reaching the assembly through `Assembly.hL2_discharged`.

- **Theorem 4.4** — the survivor analysis of Section 4 in `LemmaA.lean` (the $S4$ dead band, the valuation table, the survivor rule), the entry theorem and $B^*B = 2^{-d}I$ in `GramIdentity.lean` (`gram_upper`), the owner count and the ownership partition in `OwnerCount.lean` and `OwnerPartition.lean`, and (R1), the $a \geq b$ vanishing, as an operator identity in `OperatorBlock.lean`.

- **The assembly of Section 6** — `Assembly.lean`, with the envelope $f(e) \leq f(3) < 1$ and the row-sum bound; instantiated at the concrete T_k in `ManifestInstance.lean`.

- **The sign-blindness of Section 9** — `SignBlind.lean`. This file is a *sibling*, not a link in the chain above: it imports nothing beyond `CountingLemmas.lean`, and `gap_certificate_unconditional` neither uses nor needs it. It proves that for every odd c and every valuation pattern with $\sum j_i + 1 \leq k$ the count is $2^{k-1-\sum j_i}$, hence the probability $2^{-\sum j_i}$ with no k in it, hence identical finite-dimensional distributions for $3x+1$ and $3x-1$. What it does *not* do is construct a measure on $\mathbb{Z}_2$; see the caveat below.

Remark 8.1 (What the sign-blindness formalisation does not contain). ***`SignBlind.lean`*** *establishes the combinatorial content of the equality of the two valuation-process laws — every cylinder mass, at every depth, inside the band $\sum j_i + 1 \leq k$. It does not establish that equality as a statement about measures on $\mathbb{Z}_2$. Two steps are missing, and they are not equally hard.*

The passage from cylinder masses to a measure on $\mathbb{N}^{\mathbb{N}}$ is the lighter one, and requires no new theory: Mathlib already carries the product measure ***`Measure.infinitePiNat`*** *indexed by $\mathbb{N}$, its projective-limit property, the identification of its value on measurable cylinders with the finite product, and the independence of its coordinates. The target law is an exact fit for* ***`geometricPMF`*** *at parameter $1/2$ shifted by one, since $(1/2)^n(1/2) = 2^{-(n+1)}$. What* ***`SignBlind.lean`*** *computes* is *the cylinder mass, so the assembly is bounded work against existing library material, and no general Kolmogorov extension theorem enters.*

The harder step is tying the counting measure to Haar measure on $\mathbb{Z}_2$ — that Haar restricted to level k is uniform on residues mod 2^k, the balls being cosets of $2^k\mathbb{Z}_2$. This is standard mathematics, but Mathlib provides ***`PadicInt`*** *without the accompanying Haar development, so it is the piece that would need new material. It is avoidable at the cost of stating the result at finite level, where no $\mathbb{Z}_2$ appears at all.*

We are explicit about this because the distance between "all cylinder masses agree" and "the laws are equal" is exactly the kind of step it is tempting to elide.

Three formalisation dividends are worth recording. FACT 1, parametrised by the lift index, needs no modulo-3 argument at all. The identity $(2s^2)^{a+1} = 1$ for $s = 2^{-1/2}$ keeps every exponent in $\mathbb{N}$, so the formal assembly involves no real exponentiation. And the dead band of Lemma 4.3 must be stated in divisibility form rather than as $k - m \leq v_2(\alpha) \leq k - 2$: the valuation form is false at $\alpha = 0$ under the convention $v_2(0) = 0$, and the formalisation caught this where inspection had not.

The formalisation followed this paper's own proof of Theorem 4.4 without alteration, including the step that had been expected to be hardest. The off-diagonal entries of B^*B do not require a cancellation among roots of unity; as the proof above says, distinct columns have disjoint row support, so every off-diagonal summand already carries a zero factor.

Not formalised. Lemma C (Section 7) — it sharpens the constant rather than feeding it, so Theorem 1.1 does not depend on it — and the non-degeneracy $g_c \neq 0$, which the certificate also does not use. Nothing else in the chain of Theorem 1.1 is paper-only.

9 What this theorem does and does not say

What the theorem is a statement about

This subsection is placed first because the natural reading of Theorem 1.1 — "the mod-2^k Collatz chain mixes uniformly fast" — is wrong, and the correct reading is more interesting.

Let K_k be the *honest* 2-adic kernel: the transition matrix on odd residues mod 2^k obtained by conditioning Haar measure on the *full* 2-adic fibre of a residue, rather than on the finite lift window $[0, 2^k)$. Then K_k is not merely fast-mixing, it is **nilpotent off the Perron eigenvalue**: $|\lambda_2(K_k)| = 0$, so the chain reaches stationarity in finitely many steps. This follows from the closed form of the Bernstein–Lagarias conjugacy [3] — on the shift side each Syracuse step deletes at least one low bit, so the non-Perron part is a nilpotent shift — and is not proved here. (Numerically $|\lambda_2(K_k)|$ computed with 2^{18} lifts is $\approx 10^{-2}$ and falls steadily as the fibre is resolved, consistent with 0.)

So *the mixing of the averaged mod-2^k Collatz dynamics is trivial*, and $\mathrm{cert}(k)$ is not measuring it. What T_k differs from K_k by is exactly the substitution of the finite lift window for the full fibre, and that difference is supported on one row:

$$\mathrm{rank}(T_k - K_k) = 1 \quad \text{for every } k, \qquad \|T_k - K_k\|_1 \approx 0.25\text{–}0.50.$$

Both verified for $k = 3, \ldots, 8$. **The defect is rank one but it is not small**, and it does not shrink as k grows. A rank-one perturbation of that size can in principle drag an eigenvalue to the unit circle and destroy mixing altogether.

Theorem 1.1 says that it does not, at any scale. That is the honest content: replacing the full 2-adic fibre average by the finite lift window — which is what every computation on the mod-2^k Collatz chain actually does — perturbs a nilpotent kernel into one whose sub-dominant spectrum still sits below $0.853553\ldots$, uniformly in k, with the true values near 0.27. It is a *uniform spectral stability statement for the finite-window approximation*, and its audience is anyone computing mod-2^k Collatz statistics over a finite window and relying, usually silently, on that approximation not degenerating as the window refines.

One limit, stated plainly: this is a statement about the spectrum, not about mixing times. T_k is markedly non-normal and its eigenvector condition number grows like 10^k (Section 1, item (i)), so $|\lambda_2| \leq 0.8536$ does *not* convert into a total-variation mixing bound uniform in k. Spectrum and mixing come apart for this family. Conflating them is precisely the error of the withdrawn March 2026 draft, and it is not repaired by the present proof — only avoided, because the present proof never needs eigenvectors.

It says nothing about individual deterministic orbits.

The withdrawn cycle claim. An earlier version of this project asserted that the uniform gap eliminates non-trivial Collatz cycles. The inference is false, by a control experiment: the $3x - 1$ map (equivalently, Collatz on negative integers) has non-trivial cycles — in Syracuse form exactly $\{5, 7\}$ and $\{17, 25, 37, 55, 41, 61, 91\}$, besides the trivial $\{1\}$, and no others below 2×10^6 — yet its transfer operator passes the identical certificate, with an even stronger measured gap. The averaging over 2^k lifts that defines T_k erases exactly the deterministic orbit structure a cycle lives in, in the same way that the doubling map is mixing while having dense periodic orbits. Ruling out Collatz cycles requires instruments sensitive to specific orbits (heights, linear forms in logarithms), not the spectrum of the averaged operator.

We claim no novelty for the principle behind this control, only for the specific certificate it defeats. That the sign is invisible to this class of instrument is recorded in the literature: Kontorovich and Lagarias observe that the differences between the dynamics of $3x + 1$ and $5x + 1$ on the integers are "invisible at the level of measure theory" [5], and note that the conjugacy results of Bernstein and Lagarias [3] carry the identical ergodic theory to all $ax + b$ maps — which includes $3x - 1$. Matthews states the same phenomenon from the other direction, that for relatively prime type maps the prediction of overall cycling or divergence is independent of the remainder values [6]. That $3x - 1$ has non-trivial cycles is folklore. What is ours here is narrower and sufficient: the observation that *this* certificate, with *these* constants, is passed by $3x - 1$ with a stronger measured gap, which is what makes the withdrawn inference demonstrably false rather than merely unsupported.

Likewise the theorem has no bearing on divergent trajectories; the determinism-versus-probability gap quantified in the project's audit (a large-deviation shortfall of roughly a factor 12 for every counting-based route) remains the actual wall, consistent with the state of the art after [2].

What the theorem does provide: a natural family of arithmetically structured, non-normal, non-doubly-stochastic operators of exponentially growing dimension with a uniform, explicit, elementarily provable spectral gap, whose sharp structure ($B^*B = 2^{-d}I$ blocks, a rank-1 defect at the base-4 repunit r^*, the $31/12$ collision constant) is completely understood.

What this is good for

Three uses, none of which is progress on Collatz.

A falsification test for averaged arguments. The control experiment above generalises into a cheap and decisive check that costs a referee ten minutes. *Run the candidate argument against* $3x - 1$. If it goes through unchanged, it is wrong, because $3x - 1$ has the cycles listed above. This works because averaged models are typically blind to the sign of the shift, and that blindness is not a heuristic: for every odd c and every valuation pattern $(j_1, \ldots, j_t)$ with $\sum j_i + 1 \le k$, the number of odd residues mod 2^k realising that pattern under $x \mapsto 3x + c$ is $2^{k-1-\sum j_i}$, independent of c. Equivalently the induced probability is $2^{-\sum j_i}$, with no k in it, so the two valuation processes have identical finite-dimensional distributions. This is machine-checked (`SignBlind.lean`, sorry-free); the underlying conjugacy is classical [3, 4].

Reusable formalisation. The Lean development is public domain and modular. Independently of Collatz it supplies 2-adic fibre counts for $x \mapsto 3x + c$ mod 2^k, an orthonormal character basis on $\mathbb{Z}/2^k$ with projections graded by valuation, and a Gram identity $B^*B = 2^{-d}I$ established without forming an adjoint.

A negative result. Knowing which families of argument are closed is worth time in a problem that has consumed a great deal of it. Every route surveyed here that makes contact with the conjecture exits through the same door — an orbit-quantified statement about a hypothetical infinite trajectory — and no averaged spectral datum reaches that door.

We claim no application outside mathematics and are not aware of one.

Reproducibility

All claims are reproducible from `https://github.com/m4cd4r4/collatz-conjecture-spectral-gap`: the proof documents (`THEOREM.md` and the lemma files), the numerical ground-truth checks (`fable_assembly_check.py`, `verify_assembly.py`, `audit_halfshift_s4.py`, `lemmaB_fact1_r` the $3x-1$ control (`probe_cycle_link.py`, `CYCLE_CLAIM_REFUTED.md`), and the Lean project (`lean/`).

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